AMARNATH S. PATEL

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EDUCATION

University of Central Florida — **Undergraduate Student** — **4.00 GPA**

B.S. in Physics (Optics & Lasers) and Mathematics, Computer Science Minor — August 2025 – Present

Florida Atlantic University — **3.66 GPA**

University coursework completed via FAU High School, ages 14–18 (111 credit hours) — August 2021 – May 2025

RESEARCH EXPERIENCE

UCF Astrophotonics Lab, CREOL — Undergraduate Researcher — August 2025 – Present

Supervisor: Dr. Stephen Eikenberry

- Developing instrumentation and automation software for astrophotonics research at the intersection of photonics and observational astronomy, spanning CREOL and the UCF Physics Department.
- Built an unattended acquisition-and-reconstruction pipeline that measures the wavelength-dependent complex transfer matrix of a photonic lantern via off-axis digital holography (replicating Dobias et al., *Opt. Express* 2026); coordinates 4 instruments across all-port × C-band (1525–1575 nm) sweeps, with FFT sideband isolation, Butterworth filtering, quadratic-phase correction, and per-hologram fidelity optimization feeding LP-mode decomposition to recover complex modal amplitude and phase.
- Developing PolyOculus, control and automation software for an 8-telescope photometric observation array; implementing INDI-protocol mount control, focuser automation, RA/Dec coordinate slewing, and backlash compensation.
- Developed Python SDK wrapper (gentec-camera) for an IR beam profiling camera enabling automated acquisition, real-time beam analysis, and FITS output.

UCF Physics Department — Paid Undergraduate Research Assistant — March 2026 – Present

Physics Education Research — Supervisor: Dr. Zhongzhou Chen

- Developing ESTELA, an automated multi-version exam generation system for AI-assisted isomorphic physics problem banks, supporting scalable and equitable assessment infrastructure for introductory STEM courses.
- Built in Rust with a vanilla JS frontend; parses YAML problem banks across 13 topic areas supporting 11 question types, renders LaTeX math via KaTeX, and exports multi-version exams as LaTeX or print-ready HTML/PDF.
- Funded by NSF award 2421299 and Gates Foundation INV-076932.

FAU Grant-Funded AI Safety Research Project (High School) — January 2024 – March 2025

Supervisor: Tucker Hindle, Florida Atlantic University

- Benchmarked 10+ coherence and detection methods (BERT NSP, GPT-2 perplexity, RoBERTa, LSA, NLI, burstiness) across 3,125+ generated sequences to evaluate approaches for identifying AI-generated text.
- Identified GPT-2 perplexity as the strongest discriminator (3.3× gap between coherent and incoherent text); BERT NSP failed to distinguish the two.
- Characterized a fundamental quality–evasion tradeoff across all humanization approaches tested, including iterative sampling, content abstraction, and RL-based methods.
- Grant-funded with HPC access; findings presented at the Wilkes Honors College Symposium (2025).

OTHER EXPERIENCE

Teaching Assistant — Calculus (High School) — August 2024 – May 2025

- Assisted 70 undergraduate students with learning calculus; office hours, exam review, grading. Part-time (10 h/week).

Advanced Experimental Vehicles — Programmer, Leader, Builder (High School) — November 2023 – May 2025

- Configured Arch Linux ARM on Raspberry Pi 5 with Hyprland compositor and WireGuard VPN, enabling worldwide real-time telemetry monitoring of BMS, GPS, and camera feeds.

- Won 2nd Place in Division and Lockheed Martin Award for "Highest Level of Engineering Excellence."

PRESENTATIONS

Batch Processing for Automated Grading via Azure OpenAI June 2026

University of Central Florida, Downtown Campus.

Coherence and Detection Approaches for Identifying AI-Generated Text March 2025

Wilkes Honors College Undergraduate Research Symposium, Florida Atlantic University.

SELECTED PROJECTS

Photonic Lantern Digital Holography Automation January 2026 – Present

- Unattended pipeline measuring a photonic lantern's wavelength-dependent complex transfer matrix via off-axis digital holography. Orchestrates a tunable IR laser, fiber switch, motorized polarization controller, and GigE InGaAs camera; auto-optimizes polarization for fringe contrast, then for each port × wavelength reconstructs the complex field (FFT sideband demodulation ⟶ Butterworth filter ⟶ quadratic-phase correction) and decomposes it into LP modes — one transfer-matrix row per shot, 92% reconstruction fidelity. Ships as a single native Windows executable.

PolyOculus — 8-Telescope Array Control Software January 2026 – Present

- INDI-protocol mount control, automated focuser, RA/Dec slewing, and backlash compensation for an 8-telescope photometric observation array. Part of ongoing astrophotonics research at UCF CREOL.

ESTELA — Problem Bank Visualizer & Exam Generator March 2026 – Present

- Rust backend, vanilla JS frontend. Parses YAML problem banks across 13 topic areas and 11 question types. Renders LaTeX via KaTeX. Exports multi-version exams as LaTeX or print-ready HTML/PDF. NSF/Gates-funded research tool.

gentec-camera — IR Beam Profiling SDK November 2025

- Python SDK wrapper for Gentec IR beam profiling camera. Automated acquisition, real-time Gaussian beam analysis, FITS output for optical instrumentation research.

TECHNICAL SKILLS

Programming Languages: C/C++, Rust, Python, JavaScript, Shell (Fish, Bash, tcsh), LaTeX/Typst

Tools & Libraries: NumPy, SciPy, Docker, Git, CMake, XMake, INDI

Instrumentation: Tunable IR laser control, GigE camera acquisition, fiber switch automation, motorized polarization control, FITS data format

Operating Systems: Linux (Arch, NixOS, Ubuntu), BSD, Windows

RELEVANT COURSEWORK

University of Central Florida

Physics

Geometric Optics & Lab (A)
Chemistry Fundamentals I (A)
Quantum Information Processing (A)
Independent Research [PHY 4912] (IP)
Modern Physics [PHY 3101] (A)
Chemistry Fundamentals II (IP)
Mathematical Methods for Physics [PHZ 3113] (A)
Electricity & Magnetism I (IP)

Mathematics

Statistics [STA 2023] (A)
Linear Algebra (Proof-Based) [MAS 3106] (IP)
Partial Differential Equations (IP)
Applied Linear Algebra [MAS 3105] (A)
Complex Analysis [MAA 4402] (A)
Graph Theory (IP)

Computer Science

C Programming [EGN 3211] (A)
Object-Oriented Programming (IP)
Discrete Structures (A)

Florida Atlantic University

Physics

General Physics I (C+)
General Physics II — Honors (A)

Mathematics

Calculus I (A–)
Honors Calculus III (A)
Elementary Matrix Algebra [MAS 2103] (B)
Calculus II (A–)
Ordinary Differential Equations [MAP 2302] (A)

Computer Science

Data Structures & Algorithms (A)
Computer Architecture (A–)
C++ Programming [COP 3014] (A)
Computer Logic Design (A)
Deep Learning [CAP 4613] (A)
Web Development [COP 3813] (A)

HONORS & AWARDS

Lockheed Martin Award — "Highest Level of Engineering Excellence," AEV Competition 2024

2nd Place in Division, Advanced Experimental Vehicles Competition 2024

Florida Bright Futures — Florida Academic Scholars (highest tier; 100% tuition) 2025

NSF Award 2421299 — Research Funding (ESTELA project, with Dr. Zhongzhou Chen) 2026

Gates Foundation INV-076932 — Research Funding (ESTELA project) 2026